Week 3: Linux Logs, Sysmon, auditd and Snort Setup

Theory Notes, Developing Basic Understanding in SOC

Question 2. What is Sysmon? How does it differ from standard Windows Event Logs? Include a table comparing Sysmon Event IDs and what they monitor.

What is Sysmon

Sysmon, short for System Monitor, is a free tool from Microsoft Sysinternals. Once it is installed, it keeps running in the background and records detailed activity from the Windows machine, even after a restart.

- It runs as a service along with a driver.
- Its events are saved in Event Viewer under Applications and Services Logs, Microsoft, Windows, Sysmon, Operational.
- What it records depends on its configuration file. In the lab I use the SwiftOnSecurity config, which many SOC teams use as a starting point.

How it is different from standard Windows logs

Point	Standard Windows Event Logs	Sysmon
Process creation	Event 4688 is available only when auditing is turned on, and the command line needs an extra setting	Event ID 1 shows the full command line, hashes, parent process and user
File hashes	Not available	Hashes such as MD5 and SHA256 are recorded
Network connections	Needs extra auditing and can be very noisy	Event ID 3 shows the process with the IP address and port
DNS queries	Not logged by default	Event ID 22 shows which process asked for which domain
Parent process	Limited information	Parent process and its command line are recorded
Filtering	Basic audit policy	Detailed include and exclude rules in the config file
Availability	Built into Windows	Has to be installed separately

Important Sysmon Event IDs

Event ID	Name	What it monitors	Why it helps a SOC analyst
1	Process Create	Every new process, with its path, command line, hashes, parent and user	Helps spot suspicious commands such as encoded PowerShell
3	Network Connection	TCP and UDP connections made by a process	Helps find command and control traffic or movement to other machines

Event ID	Name	What it monitors	Why it helps a SOC analyst
5	Process Terminated	When a process ends	Helps build the timeline of a process
7	Image Loaded	DLLs loaded by a process	Helps notice DLL injection or side loading
8	CreateRemoteThread	A process creating a thread inside another process	Helps detect process injection
10	ProcessAccess	One process opening another process	Helps notice credential dumping, for example access to lsass.exe
11	File Create	A file being created or overwritten	Helps find dropped payloads and malware files
12	Registry Object Create/Delete	Registry keys or values being created or deleted	Helps notice persistence changes
13	Registry Value Set	A registry value being changed	Helps find Run key persistence and setting changes
22	DNS Query	DNS lookups along with the process that made them	Helps find suspicious domains and DNS based command and control

A few things I should remember

- Sysmon only records activity, it does not block anything. The logs need to reach a SIEM to create alerts.
- A badly written config can either create too much noise or miss important events, so tuning matters.
- The install command I use in the lab is `sysmon.exe -accepteula -i sysmonconfig-export.xml`.